C1 Geometry — Worksheet

IGCSE Mathematics 0580 | Practice Problems

C1_Q1: Angles — Problems

F Foundation [1 mark]

Q1. State the type of angle (acute, obtuse, reflex, or right angle) for an angle measuring 85^\\circ.

F Foundation [1 mark]

Q2. Find the angle x if x and 115^\\circ lie on a straight line.

F Foundation [1 mark]

Q3. Four angles meet at a point. Three of them are 72^\\circ, 108^\\circ, and 95^\\circ. Find the fourth angle.

F Foundation [1 mark]

Q4. Two straight lines intersect. One of the angles formed is 37^\\circ. Find the size of the vertically opposite angle.

C Core [1 mark]

Q5. Three angles meet at a point on a straight line. One angle is 47^\\circ, another is 68^\\circ, and the third is x. Find x.

C Core [1 mark]

Q6. Two lines intersect. Two of the angles are $3x$ and $2x + 20^\circ$, and they are vertically opposite. Find the value of x.

C Core [1 mark]

Q7. Two angles on a straight line are $(2x + 10)^\circ$ and $(3x - 20)^\circ$. Find the value of x and the size of each angle.

C Core [1 mark]

Q8. Four angles meet at a point. They are x°, $(x + 20)^\circ$, $(x + 40)^\circ$, and $(x + 60)^\circ$. Find the value of x and the size of each angle.

H Challenge [1 mark]

Q9. Two straight lines intersect. Four angles are formed: a°, b°, c°, and d°. Given that $a = 3x - 15$, $b = 2x + 25$, and a and c are vertically opposite to each other, find the values of a, b, c, and d.

H Challenge [1 mark]

Q10. Five angles meet at a point. They are x°, $2x^\circ$, $3x^\circ$, $(x + 30)^\circ$, and $(2x - 18)^\circ$. Find the value of x and the largest angle.

H Challenge [1 mark]

Q11. On a straight line AOB, $\angle AOC = (5x - 10)^\circ$, $\angle COD = (3x + 20)^\circ$, and $\angle DOB = (4x - 10)^\circ$. Find x and the measure of $\angle COD$.

H Challenge [1 mark]

Q12. Two lines intersect at O, forming vertically opposite pairs. One angle is $(5x - 7)^\circ$ and its adjacent angle is $(3x + 11)^\circ$. Find the four angles.

X Extension [1 mark]

Q13. Prove that if two adjacent angles are equal and lie on a straight line, then each is a right angle.

X Extension [1 mark]

Q14. In the diagram, AB and CD are straight lines intersecting at O, with $\angle AOC = 58^\circ$. OE bisects $\angle AOC$, and OF bisects $\angle BOD$. Find $\angle EOF$.

X Extension [1 mark]

Q15. Prove that the bisectors of two vertically opposite angles are opposite rays (i.e., lie on the same straight line).

X Extension [1 mark]

Q16. At a point O, seven rays emanate dividing the full angle into seven angles. The smallest angle is x^\\circ, and each subsequent angle is 5^\\circ more than the previous. Find x.

C1_Q2: Parallel Lines — Problems

F Foundation [1 mark]

Q17. Two parallel lines are cut by a transversal. A corresponding pair of angles measures (3x)^\\circ and 75^\\circ. Find x.

F Foundation [1 mark]

Q18. Two parallel lines are cut by a transversal. An alternate angle to a 62^\\circ angle is y^\\circ. Find y.

F Foundation [1 mark]

Q19. Two parallel lines are cut by a transversal. A pair of co-interior angles are (x)^\\circ and 110^\\circ. Find x.

F Foundation [1 mark]

Q20. In the diagram, two parallel lines are cut by a transversal. One corresponding angle is 118^\\circ. What is the size of its corresponding partner?

C Core [1 mark]

Q21. Two parallel lines are cut by a transversal. An angle is marked 47^\\circ. Find the size of the angle that is both corresponding to it and alternate to a different angle. Explain your reasoning.

C Core [1 mark]

Q22. Two parallel lines are cut by a transversal. A co-interior pair is (4x - 10)^\\circ and (2x + 40)^\\circ. Find x and both angles.

C Core [1 mark]

Q23. AB \\parallel CD. A transversal intersects them. \\angle 1 = 53^\\circ and is alternate to \\angle 2. \\angle 3 is corresponding to \\angle 1. Find \\angle 2, \\angle 3, and \\angle 4 where \\angle 4 lies on a straight line with \\angle 2.

C Core [1 mark]

Q24. Two parallel lines are cut by a transversal. One angle is 5x^\\circ and its corresponding angle is (x + 48)^\\circ. Find x and the angle measure.

H Challenge [1 mark]

Q25. AB \\parallel CD. A transversal EF cuts them. On one side of EF, the co-interior angles are (7x - 15)^\\circ and (3x + 25)^\\circ. Find x and both angles.

H Challenge [1 mark]

Q26. AB \\parallel CD. Two transversals intersect them. One transversal creates angles of 2x^\\circ and (x + 36)^\\circ as corresponding angles. The other transversal creates angles of y^\\circ and 68^\\circ as alternate angles. Find x and y.

H Challenge [1 mark]

Q27. AB \\parallel CD. A transversal meets them at P on AB and Q on CD. \\angle APQ = 4x^\\circ and \\angle PQC = 6x^\\circ. Given that \\angle APQ and \\angle PQC are co-interior, find x. Then find \\angle PQD if Q, C, D are collinear in that order.

H Challenge [1 mark]

Q28. AB \\parallel CD. A transversal creates an alternate angle pair (3x - y)^\\circ and (x + y)^\\circ, and a corresponding pair (2x + y)^\\circ and 80^\\circ. Find x and y.

X Extension [1 mark]

Q29. Prove that if a transversal cuts two lines such that co-interior angles are supplementary, then the lines are parallel.

X Extension [1 mark]

Q30. AB \\parallel CD \\parallel EF. A transversal GH intersects AB at P, CD at Q, and EF at R. \\angle APG = 48^\\circ. Find \\angle DQR, \\angle FRQ, and \\angle PQD. Give all angle reasons.

X Extension [1 mark]

Q31. In a diagram, three parallel lines are cut by two transversals. If one of the angles formed is x^\\circ and another is (180 - x)^\\circ, and they are co-interior on the same transversal, prove that x = 90.

X Extension [1 mark]

Q32. Prove that the sum of angles in any triangle is 180^\\circ using the properties of parallel lines.

C1_Q3: Polygons — Problems

F Foundation [1 mark]

Q33. Find the sum of the interior angles of a pentagon (5-sided polygon).

F Foundation [1 mark]

Q34. Find the size of each interior angle of a regular hexagon (6 sides).

F Foundation [1 mark]

Q35. Find the exterior angle of a regular octagon (8 sides).

F Foundation [1 mark]

Q36. A regular polygon has an exterior angle of 60^\\circ. How many sides does it have?

C Core [1 mark]

Q37. Four of the interior angles of a pentagon are 90^\\circ, 100^\\circ, 110^\\circ, and 120^\\circ. Find the fifth interior angle.

C Core [1 mark]

Q38. The interior angle of a regular polygon is 150^\\circ. Find the number of sides.

C Core [1 mark]

Q39. The sum of the interior angles of a polygon is 1260^\\circ. How many sides does the polygon have?

C Core [1 mark]

Q40. A regular polygon has an interior angle that is 3 times its exterior angle. Find the number of sides.

H Challenge [1 mark]

Q41. The interior angles of a hexagon are x^\\circ, (x+10)^\\circ, (x+20)^\\circ, (x+30)^\\circ, (x+40)^\\circ, and (x+50)^\\circ. Find x and the largest angle.

H Challenge [1 mark]

Q42. A regular polygon has 54 diagonals. Find the number of sides and the size of each interior angle.

H Challenge [1 mark]

Q43. The exterior angles of a polygon are in the ratio 1:2:3:4:5:6:7:8. Find the number of sides and the largest exterior angle.

H Challenge [1 mark]

Q44. The interior angle of a regular polygon exceeds its exterior angle by 140^\\circ. Find the number of sides.

X Extension [1 mark]

Q45. Prove that the sum of the exterior angles of any convex polygon is 360^\\circ, regardless of the number of sides.

X Extension [1 mark]

Q46. Two regular polygons have x and (x+5) sides respectively. The interior angle of the first polygon is 12^\\circ more than that of the second. Find x and the interior angles of both polygons.

X Extension [1 mark]

Q47. A regular polygon has n sides. The number of diagonals of the polygon is 4 times the number of sides. Find n and the interior angle.

X Extension [1 mark]

Q48. Prove that a regular polygon cannot have an interior angle of 155^\\circ. (Is it possible? If not, explain why.)

C1_Q4: Triangles — Congruence Problems

F Foundation [1 mark]

Q49. In \\triangle ABC, \\angle A = 47^\\circ and \\angle B = 63^\\circ. Find \\angle C.

F Foundation [1 mark]

Q50. In an isosceles triangle, the base angles are equal. If the vertex angle is 80^\\circ, find each base angle.

F Foundation [1 mark]

Q51. In \\triangle ABC, the exterior angle at C is 130^\\circ. If \\angle A = 55^\\circ, find \\angle B.

F Foundation [1 mark]

Q52. State which congruence condition (SSS, SAS, ASA, RHS) applies to two right-angled triangles where the hypotenuse and one side of one are equal to the hypotenuse and one side of the other.

C Core [1 mark]

Q53. In $\triangle ABC$, the angles are $2x^\circ$, $3x^\circ$, and $4x^\circ$. Find the value of x and the size of each angle.

C Core [1 mark]

Q54. In $\triangle ABC$, the exterior angle at C is 125°. $\angle A = (3x - 10)^\circ$ and $\angle B = (2x + 15)^\circ$. Find x and both angles.

C Core [1 mark]

Q55. $\triangle ABC$ is isosceles with AB = AC. $\angle B = 55^\circ$. Find $\angle A$ and $\angle C$.

C Core [1 mark]

Q56. In $\triangle ABC$ and $\triangle DEF$, AB = DE = 5 cm, $\angle B = \angle E = 40^\circ$, and BC = EF = 6 cm. Are the triangles congruent? If so, by which condition?

H Challenge [1 mark]

Q57. In $\triangle ABC$, $\angle A = 2x^\circ$, $\angle B = 3x^\circ$, and $\angle C = 4x^\circ$. $\triangle DEF$ is isosceles with $DE = DF$ and $\angle D = \angle A$. Find the base angles of $\triangle DEF$.

H Challenge [1 mark]

Q58. In quadrilateral ABCD, $AB = CD$ and $AD = BC$. Prove that $\triangle ABC \cong \triangle CDA$.

H Challenge [1 mark]

Q59. In $\triangle ABC$, $\angle ABC = 48^\circ$. The exterior angle at C is 105°. $\triangle PQR$ is such that $\angle P = \angle A$, $\angle Q = \angle B$, and $PQ = AB$. Are the triangles congruent? Explain.

H Challenge [1 mark]

Q60. In $\triangle ABC$, $AB = AC$. D is a point on BC such that AD bisects $\angle BAC$. Prove that $BD = DC$.

X Extension [1 mark]

Q61. Prove that the diagonals of a rectangle are equal using triangle congruence.

X Extension [1 mark]

Q62. In $\triangle ABC$, AB = AC. Point D lies on AB and point E lies on AC such that BD = CE. Prove that BE = CD.

X Extension [1 mark]

Q63. Prove that the exterior angle of a triangle is greater than either of the opposite interior angles, using congruence or basic angle properties.

X Extension [1 mark]

Q64. In $\triangle ABC$, AB = AC. P is the midpoint of BC. Prove that AP is perpendicular to BC.

C1_Q5: Quadrilaterals — Problems

F Foundation [1 mark]

Q65. Find the sum of the interior angles of any quadrilateral.

F Foundation [1 mark]

Q66. Three angles of a quadrilateral are 75°, 90°, and 105°. Find the fourth angle.

F Foundation [1 mark]

Q67. Name the quadrilateral that has all four sides equal and all four angles equal.

F Foundation [1 mark]

Q68. How many lines of symmetry does a rectangle (that is not a square) have?

C Core [1 mark]

Q69. The angles of a quadrilateral are x^\\circ, (2x)^\\circ, (3x)^\\circ, and (4x)^\\circ. Find x and the size of each angle.

C Core [1 mark]

Q70. In a parallelogram, one angle is 65^\\circ. Find the other three angles.

C Core [1 mark]

Q71. In an isosceles trapezium ABCD with AB \\parallel CD, \\angle A = 70^\\circ. Find the other three angles.

C Core [1 mark]

Q72. A rhombus has one angle of 120^\\circ. Find its other angles and determine how many lines of symmetry it has.

H Challenge [1 mark]

Q73. The angles of a quadrilateral are (2x+10)^\\circ, (3x-5)^\\circ, (4x+10)^\\circ, and (x-5)^\\circ. Find x and the largest angle.

H Challenge [1 mark]

Q74. In parallelogram ABCD, \\angle A = (3x+10)^\\circ and \\angle B = (2x-5)^\\circ. Find x and all four angles.

H Challenge [1 mark]

Q75. The angles of a quadrilateral are in the ratio 2:3:5:8. Find the angles and state what type of quadrilateral it is (if any specific type).

H Challenge [1 mark]

Q76. A kite has angles 80^\\circ, 80^\\circ, 130^\\circ, and x^\\circ, where x is the angle between the unequal sides. Find x and describe the symmetry of the kite.

X Extension [1 mark]

Q77. Prove that the sum of the angles of a quadrilateral is 360^\\circ by dividing it into two triangles.

X Extension [1 mark]

Q78. Prove that the diagonals of a parallelogram bisect each other.

X Extension [1 mark]

Q79. Prove that if a quadrilateral has three right angles, the fourth angle must also be a right angle.

X Extension [1 mark]

Q80. A quadrilateral has vertices at (0,0), (4,0), (5,3), and (1,3). Identify the type of quadrilateral and prove that its diagonals bisect each other.

C1_Q6: Circle Theorems — Part 1

F Foundation [1 mark]

Q81. A, B, and C lie on a circle with diameter AC. If \\angle ABC is the angle at B, find \\angle ABC.

F Foundation [1 mark]

Q82. In a circle, the angle at the centre is 80^\\circ. Find the angle subtended by the same arc at the circumference.

F Foundation [1 mark]

Q83. In a circle, two angles in the same segment are 3x^\\circ and 45^\\circ. Find x.

F Foundation [1 mark]

Q84. In a cyclic quadrilateral, one angle is 110^\\circ. Find its opposite angle.

C Core [1 mark]

Q85. A, B, C lie on a circle with diameter AC. \\angle BAC = 37^\\circ. Find \\angle BCA.

C Core [1 mark]

Q86. In a circle, the angle at the centre is (5x+10)^\\circ and the angle at the circumference subtended by the same arc is (2x+20)^\\circ. Find x and both angles.

C Core [1 mark]

Q87. In a cyclic quadrilateral, two opposite angles are 3x^\\circ and 5x^\\circ. Find x and both angles.

C Core [1 mark]

Q88. A, B, C, D lie on a circle. \\angle ACB = 34^\\circ and \\angle ABD = 42^\\circ. Find \\angle ADB and \\angle ACB if C and D are on the same side of AB.

H Challenge [1 mark]

Q89. ABCD is a cyclic quadrilateral. \\angle A = (2x+10)^\\circ, \\angle B = (3x-20)^\\circ, \\angle C = (4x)^\\circ. Find \\angle D.

H Challenge [1 mark]

Q90. AB is a diameter of a circle with centre O. C is a point on the circle such that \\angle COB = 70^\\circ. Find \\angle OCA and \\angle CAB.

H Challenge [1 mark]

Q91. ABCD is a cyclic quadrilateral. Diagonals AC and BD intersect at E. \\angle ABD = 30^\\circ, \\angle BDC = 25^\\circ. Find \\angle ACD and \\angle AEB.

H Challenge [1 mark]

Q92. O is the centre of a circle. A, B, C lie on the circle such that \\angle AOC = 120^\\circ and \\angle OAB = 20^\\circ. Find \\angle ABC.

X Extension [1 mark]

Q93. Prove that opposite angles of a cyclic quadrilateral sum to 180^\\circ.

X Extension [1 mark]

Q94. AB is a diameter of a circle with centre O. C and D are points on the circle on the same side of AB. \\angle ACD = 15^\\circ and \\angle BDC = 25^\\circ. Find \\angle CAD and \\angle CBD.

X Extension [1 mark]

Q95. In a circle, chord AB subtends \\angle ACB = 40^\\circ at the circumference. If O is the centre, find \\angle AOB. If \\angle AOB is 80^\\circ, prove the theorem that the angle at the centre is twice the angle at the circumference for this case.

X Extension [1 mark]

Q96. AB is a diameter of a circle with centre O. C and D are on the circle such that CD \\parallel AB. \\angle AOC = 50^\\circ. Find \\angle OCD, \\angle ODC, and \\angle COD.

C1_Q7: Circle Theorems — Part 2

F Foundation [1 mark]

Q97. A tangent touches a circle at point T. The radius OT is drawn to the point of contact. Find the angle between the tangent and the radius OT.

F Foundation [1 mark]

Q98. From an external point P, two tangents PA and PB are drawn to a circle. If PA = 12 cm, find PB.

F Foundation [1 mark]

Q99. A tangent at A touches a circle, and AB is a chord. The angle between the tangent and chord AB is 58°. Find the angle in the alternate segment.

F Foundation [1 mark]

Q100. A chord of length 10 cm is at a distance of 12 cm from the centre. Find the radius of the circle.

C Core [1 mark]

Q101. PA is a tangent to a circle at A, and O is the centre. OA = 6 cm and OP = 10 cm. Find the length of the tangent PA.

C Core [1 mark]

Q102. A tangent at A touches a circle. Chord AB makes an angle of 42° with the tangent. C is a point on the circle in the alternate segment. Find $\angle ACB$.

C Core [1 mark]

Q103. From external point P, tangents PA and PB touch a circle at A and B. If $\angle APB = 70^\circ$, find $\angle AOB$ where O is the centre.

C Core [1 mark]

Q104. A chord of length 24 cm is drawn in a circle of radius 13 cm. Find the distance of the chord from the centre.

H Challenge [1 mark]

Q105. TA is a tangent at A to a circle. ABC is a triangle inscribed in the circle. $\angle TAB = 55^\circ$ and $\angle ACB = 70^\circ$. Find $\angle ABC$ and $\angle BAC$.

H Challenge [1 mark]

Q106. Two tangents PA and PB are drawn to a circle from external point P. O is the centre. $\angle APB = 50^\circ$. Find $\angle AOB$ and $\angle OAB$.

H Challenge [1 mark]

Q107. A tangent at A meets chord AB at A. $\angle TAB = (3x+10)^\circ$. The angle in the alternate segment subtended by AB is $(5x-20)^\circ$. Find x and both angles.

H Challenge [1 mark]

Q108. Two parallel chords of lengths 16 cm and 12 cm lie on opposite sides of the centre of a circle of radius 10 cm. Find the distance between the chords.

X Extension [1 mark]

Q109. Prove that the tangents drawn from an external point to a circle are equal in length.

X Extension [1 mark]

Q110. Prove the alternate segment theorem: The angle between a tangent and a chord through the point of contact equals the angle in the alternate segment.

X Extension [1 mark]

Q111. Two circles of radii 5 cm and 3 cm have their centres 10 cm apart. Find the length of the common external tangent (the tangent that does not intersect the line joining the centres).

X Extension [1 mark]

Q112. Two chords AB and CD of a circle intersect at E inside the circle. Given AE = 4 cm, EB = 6 cm, CE = 3 cm, find ED. Also state the theorem used.